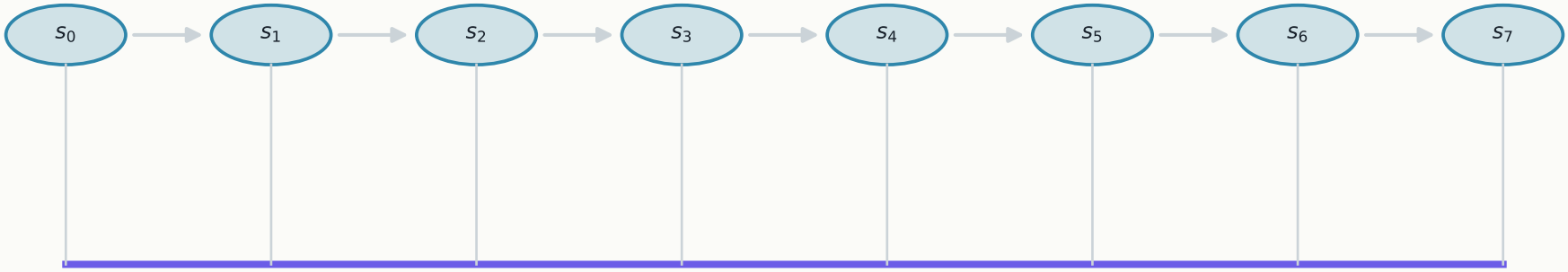
Relative correspondences propagate invertibility through a connected family



$\det T_{s_0} \neq 0$ at one fiber implies $\det T_s \neq 0$ on the whole connected base.